

NONSOFIC GROUPS EXIST

ANONYMOUS

ABSTRACT. A countable group is sofic if every finite part of its multiplication table admits arbitrarily accurate, asymptotically faithful models by permutations of finite sets. We construct an infinite, finitely presented group that is not sofic. The proof first establishes an expander–matching criterion for a property- (T) group generated by a property- (T) subgroup and finitely many elements compressing that subgroup. Median normalization, permutation conservation, and an expander Poincaré inequality convert the two component decompositions supplied by Kun’s theorem into an almost-bijective matching of components. Restriction to one matched component produces a single expanding approximation on which the Kun–Thom centralizer obstruction applies. We then verify the criterion inside

$$\mathrm{EL}_9(L_{\mathbb{F}_2}(1, 2))$$

using two explicit compressor matrices and a commuting copy of Thompson’s group V . Finally, a presentation determined by a finite portion of the multiplication table yields an infinite, finitely presented nonsocic group, without using any closure of soficity under quotients.

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1. INTRODUCTION

A countable discrete group is *sofic* if its finite multiplication tables admit arbitrarily accurate and asymptotically faithful realizations by permutations of finite sets. The approximation property arose in Gromov's work on symbolic dynamics [7]; Weiss introduced the term *sofic* and asked whether every countable group is sofic [15]. The class contains all amenable and all residually finite groups and is stable under a number of standard constructions; see, for example, [5, 14]. The question whether every group is sofic has remained open.

We give a negative answer using the binary Leavitt algebra

$$(1) \quad R = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij}, s_0 t_0 + s_1 t_1 = 1 \rangle.$$

For a unital ring A , let $\text{EL}_m(A)$ denote the subgroup of $\text{GL}_m(A)$ generated by the elementary matrices

$$x_{ij}(a) = I_m + aE_{ij}, \quad a \in A, \quad i \neq j.$$

Our first example is the Kazhdan group $\text{EL}_9(R)$. An infinite finitely presented example is then extracted from a finite part of its multiplication table.

Theorem 1.1 (Main theorem). *Let $R = L_{\mathbb{F}_2}(1, 2)$. The group*

$$G = \text{EL}_9(R)$$

is infinite, finitely generated, has property (T), and is not sofic. Moreover, there is a finite subset $F \subseteq G$ for which the group H_F defined in (50) below is infinite, finitely presented, and not sofic. In particular, not every finitely presented discrete group is sofic.

The main analytic issue is the distinction between a single expander and a disjoint union of expanders. Kun proved that a sofic approximation of a property-(T) group can be changed on a vanishing proportion of edges into a disjoint union of uniformly expanding components [10]. Kun and Thom proved that if the restriction of a sofic approximation of $\Lambda \times \Delta$ to a Kazhdan factor Λ is essentially one expander, then Δ is locally embeddable into finite groups (LEF) [11]. A disjoint union of equal-sized expanders is not enough: the commuting group may permute the components.

The Leavitt relations provide a way to pass from the disjoint union to one component. We construct a property- (T) subgroup

$$\Gamma \cong \text{EL}_8(R)$$

and two elements $u, v \in G$ such that

$$G = \langle \Gamma, u, v \rangle, \quad u\Gamma u^{-1} = v\Gamma v^{-1} =: K \leq \Gamma.$$

A hypothetical sofic approximation gives a Kun decomposition for Γ and another for G . Transporting the Γ -components by a compressor gives one-sided inclusions into other Γ -components. On each ambient G -component, normalize a Γ -component size s by

$$f = \frac{s}{s + m},$$

where m is a vertex-weighted median of the component sizes. The one-sided inclusions imply almost monotonicity of f along compressor arcs. Since each compressor is a permutation, total increase equals total decrease. The ambient expander Poincaré inequality then forces f to concentrate at its median value $1/2$. Hence the transported and target components have asymptotically equal sizes, upgrading one-sided inclusion to almost-bijection.

We restrict to one well-chosen matched component. The transported copy of Γ gives a single essentially expanding approximation of K , while every element of a commuting subgroup $J \leq \Gamma$ almost preserves the target component. Completing the restrictions to permutations produces a sofic approximation of $K \times J$ to which Kun–Thom applies.

Algebraically, J is a copy of Thompson’s group V , placed in the complementary binary corner. The group V is infinite, simple, and finitely presented [8, 4]; it is therefore not LEF. The resulting contradiction proves nonsocicity of G .

The final finite-presentation argument does not claim that soficity passes to quotients. Nonsocicity supplies a finite multiplication-table test that no finite permutation model can satisfy. Imposing exactly that finite table gives an infinite finitely presented group that retains the obstruction.

2. SOFTIC APPROXIMATIONS, EXPANDERS, AND LEF GROUPS

2.1. Sofic approximations. For a nonempty finite set Y , the normalized Hamming distance on $\text{Sym}(Y)$ is

$$d_H(\sigma, \tau) = \frac{1}{|Y|} |\{y \in Y \mid \sigma(y) \neq \tau(y)\}|.$$

Definition 2.1. A sequence of maps

$$\sigma_n : G \longrightarrow \text{Sym}(Y_n)$$

is a *sofic approximation* of a countable group G if, for all $g, h \in G$,

$$d_H(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \longrightarrow 0,$$

and for every $g \neq 1$,

$$d_H(\sigma_n(g), \text{id}_{Y_n}) \rightarrow 1.$$

The group G is *sofic* if it admits a sofic approximation.

For every fixed finite subset of G , changing each $\sigma_n(g)$ on $o(|Y_n|)$ points does not affect the approximation. We shall therefore freely make the following standard normalization.

Lemma 2.2 (Finite normalization). *Let $E = E^{-1} \subseteq G$ be finite and contain 1. After changing $\sigma_n(g)$ on $o(|Y_n|)$ points for $g \in E$, one may assume*

$$\sigma_n(1) = \text{id}, \quad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in E).$$

Proof. The sofic identities give

$$d_H(\sigma_n(1), \text{id}) \rightarrow 0 \quad \text{and} \quad d_H(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \rightarrow 0.$$

Replace the indicated permutations. Only finitely many elements are involved. $\square$

Let $S = S^{-1}$ be a finite generating set. The S -generator graph associated with σ_n has vertex set Y_n and a directed arc

$$y \xrightarrow{s} \sigma_n(s)y$$

for each $s \in S$. After the normalization above, forgetting orientation and labels gives a bounded-degree undirected multigraph. All edge-counting estimates below are insensitive to loops, multiplicities, and bounded factors coming from orientation.

2.2. Expanders and essential expansion. For a finite graph X and $A \subseteq V(X)$, let $\partial_X A$ be the set of edges with one endpoint in A and one outside. The Cheeger constant is

$$h(X) = \min_{0 < |A| \leq |V(X)|/2} \frac{|\partial_X A|}{|A|}.$$

A bounded-degree sequence (X_n) is an *expander sequence* if $|V(X_n)| \rightarrow \infty$ and $\inf_n h(X_n) > 0$.

We say that two bounded-degree graph sequences on the same vertex sets are *essentially equal* if the symmetric difference of their edge multisets has size $o(|V(X_n)|)$. A sequence is an *essential expander sequence* if it is essentially equal to an expander sequence.

We use the following form of Kun's theorem.

Theorem 2.3 (Kun [10]). *Let Λ be a finitely generated property-(T) group and let S be a finite symmetric generating set. For every sofic approximation of Λ , the associated S -generator graphs are essentially equal to graphs whose connected components have Cheeger constants bounded below by one positive constant independent of the component and of n .*

Uniform Cheeger expansion gives a uniform Poincaré inequality. We record only the form needed here.

Lemma 2.4 (Poincaré inequality). *For every degree bound d and $h > 0$, there is $C = C(d, h)$ such that the following holds. If X has maximum degree at most d and $h(X) \geq h$, then every $f : V(X) \rightarrow \mathbb{R}$ satisfies*

$$\sum_{x \in V(X)} |f(x) - \bar{f}_X|^2 \leq C \sum_{\{x, y\} \in E(X)} |f(x) - f(y)|^2,$$

where $\bar{f}_X = |V(X)|^{-1} \sum_x f(x)$.

Proof. This is the standard Cheeger inequality relating the first nonzero eigenvalue of the normalized Laplacian to the Cheeger constant, followed by the spectral Poincaré inequality. $\square$

2.3. LEF groups and the Kun–Thom obstruction.

Definition 2.5. A group D is *locally embeddable into finite groups*, abbreviated *LEF*, if for every finite $F \subseteq D$ there are a finite group H and an injective map $\phi : F \rightarrow H$ such that

$$\phi(x)\phi(y) = \phi(xy)$$

whenever $x, y, xy \in F$.

Lemma 2.6. *Every finitely presented LEF group is residually finite.*

Proof. Let $D = \langle S \mid \mathcal{R} \rangle$ be finitely presented and let $1 \neq g \in D$. Choose a word w in $S^{\pm 1}$ representing g . Let $F \subseteq D$ contain all elements represented by prefixes of the finitely many relators and their cyclic products, all elements represented by prefixes of w , and all products needed to read those words one letter at a time. A local embedding $\phi : F \rightarrow H$ into a finite group sends the generators to elements satisfying every relator. It therefore extends to a homomorphism $D \rightarrow H$. Injectivity of ϕ on F ensures that the image of g is nontrivial. Thus every nonidentity element survives in a finite quotient. $\square$

The following is the centralizer obstruction of Kun and Thom. Their proof applies unchanged when the induced graph is only essentially an expander; equivalently, one may absorb the vanishing edge modification into the approximation errors.

Theorem 2.7 (Kun–Thom [11]). *Let Λ be a countable property-(T) group and let D be finitely generated. Suppose that $\Lambda \times D$ has a sofic approximation for which the edges labelled by a fixed finite generating set of Λ form an essential expander sequence. Then D is LEF.*

3. AN EXPANDER–MATCHING CRITERION

We now prove the analytic criterion used in the construction.

Theorem 3.1 (Expander–matching criterion). *Let G be a finitely generated property-(T) group, and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set $Q \subseteq G$ such that*

$$(2) \quad G = \langle \Gamma, Q \rangle$$

and

$$(3) \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$ and put

$$K = q_0\Gamma q_0^{-1}.$$

Let $J \leq \Gamma$ be finitely generated and suppose

$$(4) \quad [K, J] = 1, \quad K \cap J = \{1\}.$$

If G is sofic, then J is LEF.

Throughout the proof, write $N_n = |Y_n|$, and use $o(N_n)$ for quantities whose ratio to N_n tends to zero. All implicit constants depend only on fixed generating sets.

3.1. The two component decompositions. Assume that

$$\sigma_n : G \longrightarrow \text{Sym}(Y_n)$$

is a sofic approximation. Choose a finite symmetric generating set $S = S^{-1}$ for Γ , and enlarge Q , if necessary, so that

$$A = S \cup Q \cup Q^{-1}$$

is a finite symmetric generating set for G . Apply Lemma 2.2 to the finite set needed below.

Apply Theorem 2.3 to the restriction to Γ . We obtain a graph $\tilde{X}_n$ on Y_n , differing from the S -generator graph in $o(N_n)$ edges, whose connected components have Cheeger constant at least $h_\Gamma > 0$. Let

$$\mathcal{P}_n$$

be its component partition, and let $P_n(y)$ denote the block containing y .

Apply Theorem 2.3 to G . We obtain a graph $\tilde{Z}_n$, differing from the A -generator graph in $o(N_n)$ edges, whose components have Cheeger constant at least $h_G > 0$. Let

$$\mathcal{B}_n$$

be its component partition, and write $B_n(y)$ for the block containing y .

For every fixed $\gamma \in \Gamma$,

$$(5) \quad |\{y \in Y_n \mid P_n(\sigma_n(\gamma)y) \neq P_n(y)\}| = o(N_n).$$

Indeed, this is immediate for $\gamma \in S$ from the edge modification, and for a fixed S -word it follows by telescoping the word and adding the sofic multiplication defects. Similarly, for every $a \in A$,

$$(6) \quad |\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}| = o(N_n).$$

3.2. A refinement estimate.

Lemma 3.2 (Expander refinement). *Let X be a finite graph with $h(X) \geq h > 0$, and let $\mathcal{R}$ be a partition of $V(X)$. Let $R_X \in \mathcal{R}$ maximize $|V(X) \cap R|$, and let $e_{\mathcal{R}}(X)$ be the number of edges of X whose endpoints lie in different $\mathcal{R}$ -blocks. Then*

$$|V(X) \setminus R_X| \leq \frac{2}{h} e_{\mathcal{R}}(X).$$

Proof. Put $D = V(X)$ and $r = |D \setminus R_X|$. If $r \leq |D|/2$, expansion of $D \setminus R_X$ gives at least hr crossing edges.

Suppose $r > |D|/2$. Then every $\mathcal{R}$ -piece contained in D has size at most $|D|/2$. Summing the Cheeger inequality over all such pieces gives at least $h|D|$ boundary incidences. Each crossing edge is counted twice, so

$$e_{\mathcal{R}}(X) \geq \frac{h|D|}{2} \geq \frac{hr}{2}.$$

Both cases give the assertion. $\square$

Fix $q \in Q$. For each $C \in \mathcal{P}_n$, transport the expander $\tilde{X}_n[C]$ by $\sigma_n(q)$. This gives an h_{Γ} -expander on $\sigma_n(q)C$. Apart from $o(N_n)$ total edges, its edges agree with edges labelled by the fixed elements

$$qSq^{-1} \subseteq \Gamma.$$

Indeed, the original Kun modifications contribute $o(N_n)$ edges, and for each $s \in S$, sofic multiplicativity gives

$$(7) \quad \sigma_n(q)\sigma_n(s)\sigma_n(q)^{-1} \sim \sigma_n(qsq^{-1})$$

with Hamming error $o(1)$.

By (5), only $o(N_n)$ of the transported edges cross the partition $\mathcal{P}_n$. Lemma 3.2, summed over the transported components, gives a map

$$\Phi_{q,n} : \mathcal{P}_n \longrightarrow \mathcal{P}_n$$

such that, with

$$(8) \quad e_{q,n}(C) = |\sigma_n(q)C \setminus \Phi_{q,n}(C)|,$$

we have

$$(9) \quad \sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n).$$

Thus each transported component is almost contained, in a global weighted sense, in one original Γ -component.

3.3. Median normalization and conservation. Define

$$s_n(y) = |P_n(y)|.$$

For every $B \in \mathcal{B}_n$, choose a vertex-weighted median $m_n(B)$ of the values $s_n(y)$, $y \in B$: at least half the vertices of B satisfy $s_n(y) \leq m_n(B)$, and at least half satisfy $s_n(y) \geq m_n(B)$. Define

$$(10) \quad f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))}.$$

Then $0 < f_n < 1$, and $1/2$ is a median of f_n on every $B \in \mathcal{B}_n$.

For $s \in S$, if both the s -arc stays in the same $\mathcal{P}_n$ -block and in the same $\mathcal{B}_n$ -block, then f_n is unchanged. Equations (5) and (6) therefore imply

$$(11) \quad \sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n) \quad (s \in S).$$

Now fix $q \in Q$. Let $C = P_n(y)$ and $D = \Phi_{q,n}(C)$. Suppose

$$(12) \quad \sigma_n(q)y \in D \quad \text{and} \quad B_n(\sigma_n(q)y) = B_n(y) = B.$$

Since $|\sigma_n(q)C| = |C|$,

$$|D| \geq |C| - e_{q,n}(C).$$

For $a, m > 0$ and $0 \leq \delta \leq 1$,

$$(13) \quad \frac{a}{a+m} - \frac{(1-\delta)a}{(1-\delta)a+m} \leq \delta.$$

Applying this with $a = |C|$, $m = m_n(B)$, and $\delta = e_{q,n}(C)/|C|$, we obtain on the points satisfying (12)

$$(f_n(y) - f_n(\sigma_n(q)y))_+ \leq \frac{e_{q,n}(C)}{|C|}.$$

Summing first over each C , and adding the exceptional points in (8) and the q -arcs crossing $\mathcal{B}_n$, gives

$$(14) \quad \sum_y (f_n(y) - f_n(\sigma_n(q)y))_+ = o(N_n).$$

Permutation conservation is exact:

$$(15) \quad \sum_y f_n(\sigma_n(q)y) = \sum_y f_n(y).$$

Consequently the total positive and negative variations are equal, and (14) implies

$$(16) \quad \sum_y |f_n(\sigma_n(q)y) - f_n(y)| = o(N_n) \quad (q \in Q \cup Q^{-1}).$$

Since $|r|^2 \leq |r|$ for $|r| \leq 1$, equations (11) and (16) show that the Dirichlet energy of f_n in the A -generator graph is $o(N_n)$. Changing $o(N_n)$ edges does

not affect that conclusion. Lemma 2.4, applied on every component of $\tilde{Z}_n$, yields

$$(17) \quad \sum_{B \in \mathcal{B}_n} \sum_{y \in B} |f_n(y) - \bar{f}_n(B)|^2 = o(N_n),$$

where

$$\bar{f}_n(B) = \frac{1}{|B|} \sum_{y \in B} f_n(y).$$

Because $1/2$ is a median of $f_n|_B$,

$$(18) \quad |B| |\bar{f}_n(B) - \tfrac{1}{2}|^2 \leq 2 \sum_{y \in B} |f_n(y) - \bar{f}_n(B)|^2.$$

Using the orthogonal decomposition about the mean, we conclude

$$(19) \quad \frac{1}{N_n} \sum_{y \in Y_n} |f_n(y) - \tfrac{1}{2}|^2 \longrightarrow 0.$$

The normalized constant $1/2$ means precisely that component sizes concentrate at the median:

$$\frac{s}{s+m} \rightarrow \frac{1}{2} \iff \frac{s}{m} \rightarrow 1.$$

3.4. Almost-bijective matching.

Proposition 3.3 (Component matching). *For every fixed $q \in Q$, there are collections $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$ such that*

- (i) $\sum_{C \in \mathcal{G}_{q,n}} |C| = (1 - o_n(1))N_n$;
 - (ii) *uniformly for $C \in \mathcal{G}_{q,n}$,*
- $$(20) \quad |\sigma_n(q)C \triangle \Phi_{q,n}(C)| = o_n(1)|C|;$$
- (iii) *the restriction of $\Phi_{q,n}$ to $\mathcal{G}_{q,n}$ is injective;*
 - (iv) *the distinct target blocks have total mass $(1 - o_n(1))N_n$.*

Proof. Choose $\eta_n \downarrow 0$ sufficiently slowly that all global errors in (6), (9), and (19) are $o(\eta_n^4 N_n)$. Let W_n be the union of the following bad vertex sets:

- points y with $|f_n(y) - 1/2| > \eta_n$;
- points y with $|f_n(\sigma_n(q)y) - 1/2| > \eta_n$;
- points whose q -arc crosses $\mathcal{B}_n$;
- points $y \in C$ for which $\sigma_n(q)y \notin \Phi_{q,n}(C)$.

Equations already proved give $|W_n| = o(\eta_n^2 N_n)$. Discard every source block C for which

$$e_{q,n}(C) > \eta_n |C| \quad \text{or} \quad |C \cap W_n| > \eta_n |C|.$$

The discarded blocks have total mass $o(N_n)$. Call the remaining collection $\mathcal{G}_{q,n}$.

For $C \in \mathcal{G}_{q,n}$, choose $y \in C \setminus W_n$, put $D = \Phi_{q,n}(C)$, and let $B = B_n(y) = B_n(\sigma_n(q)y)$. From (10),

$$|f_n(y) - \tfrac{1}{2}| \leq \eta_n \implies \frac{1 - 2\eta_n}{1 + 2\eta_n} \leq \frac{|C|}{m_n(B)} \leq \frac{1 + 2\eta_n}{1 - 2\eta_n}.$$

The same estimate for $\sigma_n(q)y \in D$ gives the identical bound with $|D|$ in place of $|C|$. Hence

$$(21) \quad |D| = (1 + O(\eta_n))|C|.$$

Together with $e_{q,n}(C) \leq \eta_n|C|$, this gives

$$\begin{aligned} |\sigma_n(q)C \triangle D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\leq 2e_{q,n}(C) + ||D| - |C|| \\ &= O(\eta_n)|C|. \end{aligned}$$

This proves (i) and (ii).

For n large, two distinct good source blocks cannot have the same target. Their images under the bijection $\sigma_n(q)$ are disjoint, while by (20) and (21) each would occupy a $(1 - o_n(1))$ -fraction of the same target block. Thus $\Phi_{q,n}$ is injective on $\mathcal{G}_{q,n}$. Finally,

$$\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 - o_n(1)) \sum_{C \in \mathcal{G}_{q,n}} |C| = (1 - o_n(1))N_n,$$

which proves (iv). $\square$

3.5. Selection of one component. We need a standard weighted diagonal selection. We state it in the form used below.

Lemma 3.4 (Weighted component selection). *For each n , let*

$$\{(C_{n,i}, D_{n,i}) : i \in I_n\}$$

be pairs of subsets of a finite set Y_n . Assume that the $C_{n,i}$ are pairwise disjoint, the $D_{n,i}$ are pairwise disjoint, and

$$\sum_i |D_{n,i}| = (1 - o_n(1))|Y_n|.$$

For each $r \in \mathbb{N}$, suppose nonnegative errors $E_{n,i}^{(r)}$ satisfy

$$\sum_i E_{n,i}^{(r)} = o(|Y_n|)$$

for fixed r . Suppose also that, for every M , the total mass of the $D_{n,i}$ with $|D_{n,i}| \leq M$ is $o(|Y_n|)$. Then there are indices $i(n)$ such that

$$|D_{n,i(n)}| \longrightarrow \infty$$

and, for every fixed r ,

$$E_{n,i(n)}^{(r)} = o(|D_{n,i(n)}|).$$

Proof. Choose $r_n \rightarrow \infty$ so slowly that

$$\sum_i \sum_{r \leq r_n} E_{n,i}^{(r)} = o(|Y_n|).$$

Choose $\delta_n \downarrow 0$ so slowly that the left side is $o(\delta_n |Y_n|)$. The total mass of indices for which

$$\sum_{r \leq r_n} E_{n,i}^{(r)} > \delta_n |D_{n,i}|$$

is $o(|Y_n|)$. Discard these indices and also those with $|D_{n,i}| \leq r_n$. Some index remains, and any remaining index has the required properties. $\square$

We also verify the size hypothesis in our situation.

Lemma 3.5 (Component sizes diverge in mass). *For every fixed M , the union of the $\mathcal{P}_n$ -blocks of size at most M has size $o(N_n)$.*

Proof. Choose a finite set $E \subseteq \Gamma$ with $|E| > M$. By asymptotic faithfulness, for all but $o(N_n)$ points y , the points $\sigma_n(g)y$, $g \in E$, are pairwise distinct. By (5), for all but $o(N_n)$ points they all lie in $P_n(y)$. Hence all but $o(N_n)$ points lie in blocks of size at least $|E| > M$. $\square$

3.6. Proof of the criterion.

Proof of Theorem 3.1. Apply Proposition 3.3 with $q = q_0$. Write the good matched pairs as

$$(C, D), \quad D = \Phi_{q_0,n}(C).$$

The source blocks are disjoint, the target blocks are disjoint, and the targets have total mass $(1 - o_n(1))N_n$.

Let

$$L_1 \subseteq L_2 \subseteq \cdots \subseteq KJ$$

be a finite symmetric exhaustion containing 1, arranged so that products needed to test multiplication on L_r lie in L_{r+1} . For each matched pair and each fixed r , define an error $E_{n,C,D}^{(r)}$ to be the sum of:

- (a) the matching error $|\sigma_n(q_0)C \triangle D|$;
- (b) the number of Kun-modified S -edges incident to C ;
- (c) the conjugacy defects in (7), restricted to C , for $s \in S$;
- (d) the multiplication defects for pairs in L_r , restricted to D ;
- (e) for $1 \neq g \in L_r$, the number of points of D fixed by $\sigma_n(g)$;
- (f) for $g \in L_r$, the number of points of D sent outside D by $\sigma_n(g)$.

For fixed r , the sum of these errors over all matched pairs is $o(N_n)$. The source blocks are disjoint for (b) and (c), the target blocks are disjoint for (d)–(f), soficity handles (c)–(e), and (5) handles (f), because

$$KJ \leq \Gamma.$$

The matching estimate handles (a). Lemmas 3.4 and 3.5 therefore give matched pairs

$$(22) \quad (C_n, D_n), \quad D_n = \Phi_{q_0,n}(C_n),$$

such that

$$(23) \quad |D_n| \rightarrow \infty$$

and every fixed finite family of the errors above is $o(|D_n|)$.

Put

$$U_n = \sigma_n(q_0)C_n.$$

We know

$$(24) \quad |U_n \triangle D_n| = o(|D_n|).$$

The sets need not have exactly the same cardinality. Choose $D'_n \subseteq Y_n$ such that

$$(25) \quad |D'_n| = |U_n| \quad \text{and} \quad |D'_n \triangle D_n| \leq |U_n \triangle D_n|.$$

If $|D_n| > |U_n|$, remove the required number of points from $D_n \setminus U_n$; if $|D_n| < |U_n|$, add points from $U_n \setminus D_n$. Every selected local estimate transfers from D_n to D'_n , since changing a set by r points changes each fixed permutation escape count, fixed-point count, or multiplication-defect count by at most $O(r)$.

For $g \in KJ$, the restriction of $\sigma_n(g)$ to

$$D'_n \cap \sigma_n(g)^{-1}D'_n$$

is a bijection onto $D'_n \cap \sigma_n(g)D'_n$, and its complement has size $o(|D'_n|)$. Complete this partial bijection arbitrarily to a permutation

$$\pi_n(g) \in \text{Sym}(D'_n).$$

The completion changes $\sigma_n(g)$ on only $o(|D'_n|)$ points for each fixed g . The selected multiplication and fixed-point estimates imply that

$$\pi_n : KJ \longrightarrow \text{Sym}(D'_n)$$

is a sofic approximation. Since $[K, J] = 1$ and $K \cap J = 1$,

$$KJ \cong K \times J.$$

It remains to show that the K -generator graph is essentially one expander. Let

$$S_K = q_0 S q_0^{-1}.$$

Transport the expander $\tilde{X}_n[C_n]$ by $\sigma_n(q_0)$, obtaining an h_Γ -expander on U_n . Because $|U_n| = |D'_n|$, choose a bijection

$$\theta_n : U_n \longrightarrow D'_n$$

which fixes $U_n \cap D'_n$. The number of moved vertices is $o(|D'_n|)$ by (24) and (25). Push the transported expander through θ_n . This remains an h_Γ -expander on D'_n .

The pushed graph differs from the S_K -generator graph of π_n in only $o(|D'_n|)$ edges. The errors come from the original Kun modifications in C_n , the conjugacy defects

$$\sigma_n(q_0)\sigma_n(s)\sigma_n(q_0)^{-1} \sim \sigma_n(q_0 s q_0^{-1}),$$

the vertices moved by θ_n , and the completion of the partial restrictions. All were included in the selected error budget.

Thus the K -edges form an essential expander sequence. The group $K \cong \Gamma$ has property (T). Theorem 2.7, applied to $K \times J$, now implies that J is LEF. $\square$

Remark 3.6. The argument never claims that the ambient generators fix every Γ -component. They may permute equal-sized components, as in a product approximation $X_n \times Z_n$. The proof instead chooses one matched target component which is almost preserved by $KJ \leq \Gamma$, while the transported source component supplies expansion there. This is the point that avoids the standard product obstruction.

4. THE BINARY LEAVITT ALGEBRA AND ITS ELEMENTARY GROUPS

Let $R = L_{\mathbb{F}_2}(1, 2)$ be given by (1). Put

$$p_0 = s_0 t_0, \quad p_1 = s_1 t_1.$$

Then p_0, p_1 are orthogonal idempotents with $p_0 + p_1 = 1$.

For a word $\alpha = s_{i_1} \cdots s_{i_r}$, set

$$\alpha^* = t_{i_r} \cdots t_{i_1}.$$

If α and β are prefix-incomparable, then $\alpha^* \beta = 0$, while $\alpha^* \alpha = 1$.

4.1. The equality $\text{GL}_m(R) = \text{EL}_m(R)$. We require that the explicit invertible matrices below are elementary. The following fact is proved in [9, Proposition 4.2]; we include the argument.

Proposition 4.1. *For $R = L_{\mathbb{F}_2}(1, 2)$,*

$$K_1(R) = 0, \quad R^\times = [R^\times, R^\times], \quad \text{GL}_m(R) = \text{EL}_m(R) \quad (m \geq 2).$$

Proof. The binary rose Leavitt path algebra is a unital purely infinite simple ring [1]. Purely infinite simple rings are GE-rings, and the natural map from the unit group induces

$$(R^\times)_{\text{ab}} \cong K_1(R)$$

[3, Theorem 2.4 and its proof].

The graph K -theory sequence of Ara–Brustenga–Cortiñas [2, Theorem 7.6], applied to the one-vertex two-loop graph, contains

$$K_1(\mathbb{F}_2) \xrightarrow{-1} K_1(\mathbb{F}_2) \longrightarrow K_1(R) \longrightarrow K_0(\mathbb{F}_2) \xrightarrow{-1} K_0(\mathbb{F}_2).$$

The first map is inversion on $\mathbb{F}_2^\times = \{1\}$, and the last is multiplication by -1 on $\mathbb{Z}$; both are isomorphisms. Hence $K_1(R) = 0$, so $R^\times$ is perfect.

For completeness, if $c \in [R^\times, R^\times]$, then

$$\text{diag}(c, 1) \in \text{EL}_2(R).$$

Indeed, for $a \in R^\times$,

$$w(a) = x_{12}(a)x_{21}(-a^{-1})x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix},$$

so $w(a)w(-1) = \text{diag}(a, a^{-1})$. A product of three such diagonal matrices expresses $\text{diag}([a, b], 1)$ as an elementary matrix product. Thus every invertible diagonal matrix belongs to $\text{EL}_m(R)$. Since R is a GE-ring, $\text{GL}_m(R) = \text{EL}_m(R)$ for every $m \geq 2$. $\square$

The ring R is finitely generated as an associative unital ring. We shall use the following theorem.

Theorem 4.2 (Ershov–Jaikin-Zapirain [6]). *If A is a finitely generated associative unital ring and $m \geq 3$, then $\text{EL}_m(A)$ has property (T).*

Set

$$(26) \quad G = \text{EL}_9(R)$$

and let

$$(27) \quad \Gamma = \left\{ \begin{pmatrix} \gamma & 0 \\ 0 & 1 \end{pmatrix} \mid \gamma \in \text{EL}_8(R) \right\}.$$

By Theorem 4.2, both G and $\Gamma \cong \text{EL}_8(R)$ have property (T).

5. TWO COMPRESSORS WITH A COMMON IMAGE

5.1. A nine-leaf binary partition. Define

$$(28) \quad \alpha_i = s_0^{i-1} s_1 \quad (1 \leq i \leq 8), \quad \alpha_9 = s_0^8.$$

These are the leaves obtained by repeatedly expanding the leftmost leaf in the binary rooted tree. They satisfy

$$(29) \quad \alpha_i^* \alpha_j = \delta_{ij}$$

and

$$(30) \quad \sum_{i=1}^9 \alpha_i \alpha_i^* = 1.$$

For (30), telescope:

$$\begin{aligned} \sum_{i=1}^8 s_0^{i-1} s_1 t_1 t_0^{i-1} + s_0^8 t_0^8 &= \sum_{j=0}^7 s_0^j (1 - s_0 t_0) t_0^j + s_0^8 t_0^8 \\ &= \sum_{j=0}^7 (s_0^j t_0^j - s_0^{j+1} t_0^{j+1}) + s_0^8 t_0^8 \\ &= 1. \end{aligned}$$

5.2. The first compressor. Let

$$\mathbf{b} = \begin{pmatrix} s_1 \alpha_1^* \\ \vdots \\ s_1 \alpha_8^* \end{pmatrix}, \quad \mathbf{c} = (\alpha_1 t_1 \quad \cdots \quad \alpha_8 t_1).$$

Define

$$(31) \quad u = \begin{pmatrix} s_0 I_8 & \mathbf{b} \\ 0 & \alpha_9^* \end{pmatrix}.$$

Lemma 5.1. *The matrix u is invertible, with*

$$(32) \quad u^{-1} = \begin{pmatrix} t_0 I_8 & 0 \\ \mathbf{c} & \alpha_9 \end{pmatrix}.$$

In particular, $u \in G$.

Proof. Using (29),

$$(\mathbf{bc})_{ij} = s_1 \alpha_i^* \alpha_j t_1 = \delta_{ij} p_1.$$

Moreover,

$$\mathbf{b} \alpha_9 = 0, \quad \alpha_9^* \mathbf{c} = 0, \quad \mathbf{cb} = \sum_{i=1}^8 \alpha_i \alpha_i^* = 1 - \alpha_9 \alpha_9^*.$$

Together with $s_0 t_0 + p_1 = 1$, $t_0 s_0 = 1$, and (30), these identities show that the matrices in (31) and (32) multiply to I_9 in both orders. Hence $u \in \text{GL}_9(R) = \text{EL}_9(R) = G$ by Proposition 4.1. $\square$

For $\gamma = (\gamma_{ij}) \in M_8(R)$, write $s_0 \gamma t_0$ for the matrix with entries $s_0 \gamma_{ij} t_0$.

Proposition 5.2. *For $\gamma \in \text{EL}_8(R)$,*

$$(33) \quad u \begin{pmatrix} \gamma & 0 \\ 0 & 1 \end{pmatrix} u^{-1} = \begin{pmatrix} p_1 I_8 + s_0 \gamma t_0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Consequently,

$$(34) \quad K := u \Gamma u^{-1} = \langle x_{ij}(s_0 a t_0) \mid a \in R, 1 \leq i \neq j \leq 8 \rangle \leq \Gamma.$$

Proof. Multiplying the block matrices gives top-left block

$$s_0 \gamma t_0 + \mathbf{bc} = s_0 \gamma t_0 + p_1 I_8.$$

The off-diagonal blocks vanish by prefix orthogonality, and the bottom-right entry is 1. For an elementary generator $\gamma = I_8 + a E_{ij}$, the top-left block becomes

$$p_1 I_8 + s_0 I_8 t_0 + s_0 a t_0 E_{ij} = I_8 + s_0 a t_0 E_{ij}.$$

This proves the description of K , which lies in the upper-left $\text{EL}_8(R)$ -copy. $\square$

5.3. The second compressor and generation of G . Let $z \in M_9(R)$ act as the identity on coordinates $2, \dots, 8$, and let its block on coordinates $1, 9$ be

$$(35) \quad \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}.$$

Lemma 5.3. *The matrix z is an involution in G , and it centralizes K .*

Proof. The square of the block in (35) is

$$\begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus $z^2 = 1$, so $z \in \text{GL}_9(R) = G$.

It is enough to check the elementary generators of K . Put $r = s_0 a t_0$. Then

$$p_0 r = r, \quad t_1 r = 0, \quad r p_0 = r, \quad r s_1 = 0.$$

Direct multiplication gives

$$z x_{1j}(r) z = x_{1j}(r), \quad z x_{j1}(r) z = x_{j1}(r) \quad (2 \leq j \leq 8),$$

and generators not involving the first coordinate are fixed trivially. $\square$

Set

$$(36) \quad v = zu.$$

Then

$$(37) \quad v \Gamma v^{-1} = z K z^{-1} = K = u \Gamma u^{-1}.$$

Proposition 5.4. *The ambient group is generated by the subgroup and the two compressors:*

$$(38) \quad G = \langle \Gamma, u, v \rangle.$$

Proof. Since $z = v u^{-1}$, it suffices to prove $\langle \Gamma, z \rangle = G$. For $2 \leq j \leq 8$ and $a \in R$, direct multiplication with the block (35) gives

$$(39) \quad z x_{1j}(s_1 a) z = x_{9j}(a)$$

and

$$(40) \quad z x_{j1}(a t_1) z = x_{j9}(a).$$

Thus $\langle \Gamma, z \rangle$ contains every elementary matrix $x_{9j}(a)$ and $x_{j9}(a)$ with $2 \leq j \leq 8$. The Steinberg commutator identities yield

$$[x_{12}(a), x_{29}(1)] = x_{19}(a), \quad [x_{92}(a), x_{21}(1)] = x_{91}(a).$$

Hence all elementary matrices involving the ninth coordinate lie in $\langle \Gamma, z \rangle$, as do all elementary matrices on the first eight coordinates. These generate $\text{EL}_9(R)$. $\square$

6. A COMMUTING COPY OF THOMPSON'S GROUP V

A finite binary tree table

$$\beta_i \xi \mapsto \alpha_i \xi$$

defines an element of Thompson's group V . The assignment

$$(41) \quad g \mapsto U_g = \sum_i \alpha_i \beta_i^*$$

embeds V into $R^\times$; multiplication follows after passing to a common refinement of tree tables. This is the standard Leavitt-algebra realization of the Higman–Thompson group; see [13, 9].

For $g \in V \leq R^\times$, define

$$(42) \quad c(g) = p_0 + s_1 g t_1.$$

Lemma 6.1. *The map $c : V \rightarrow R^\times$ is an injective homomorphism. Moreover,*

$$(43) \quad c(g)s_0 = s_0, \quad t_0 c(g) = t_0.$$

Proof. Orthogonality of the two corners gives

$$c(g)c(h) = p_0 + s_1 g h t_1 = c(gh)$$

and $c(g)^{-1} = c(g^{-1})$. Also

$$t_1 c(g) s_1 = g,$$

so the map is injective. Finally,

$$c(g)s_0 = p_0 s_0 + s_1 g t_1 s_0 = s_0,$$

and similarly $t_0 c(g) = t_0$. □

Define

$$(44) \quad j(g) = \text{diag}(c(g), 1, \dots, 1) \in \text{GL}_9(R),$$

where the final coordinate is the ninth coordinate and the upper-left 8×8 block is $\text{diag}(c(g), 1, \dots, 1)$. Proposition 4.1 gives

$$\text{diag}(c(g), 1, \dots, 1) \in \text{EL}_8(R),$$

so

$$(45) \quad J := j(V) \leq \Gamma.$$

Proposition 6.2. *The subgroups K and J satisfy*

$$[K, J] = 1 \quad \text{and} \quad K \cap J = \{1\}.$$

Consequently,

$$KJ \cong K \times J.$$

Proof. Let $r = s_0 a t_0$. Equation (43), applied also to g^{-1} , gives

$$c(g)r = r, \quad rc(g)^{-1} = r.$$

Thus conjugation by $j(g)$ fixes every generator $x_{ij}(r)$ of K , proving $[K, J] = 1$.

By (33), every $k \in K$ has upper-left block

$$p_1 I_8 + s_0 \gamma t_0$$

for some $\gamma \in \text{EL}_8(R)$. Hence

$$(46) \quad t_1(k_{11} - 1)s_1 = 0.$$

On the other hand,

$$(47) \quad t_1(c(g) - 1)s_1 = g - 1.$$

If $j(g) \in K$, equations (46) and (47) imply $g = 1$. Thus the intersection is trivial. $\square$

Proposition 6.3. *The group $J \cong V$ is not LEF.*

Proof. Thompson's group V is infinite, simple, and finitely presented [8, 4]. By Lemma 2.6, a finitely presented LEF group is residually finite. An infinite simple group has no nontrivial finite quotient: the kernel of such a quotient would be a proper normal subgroup, while an injective map into a finite group is impossible. Hence V , and therefore J , is not LEF. $\square$

7. NONSOFTICITY OF THE ELEMENTARY GROUP

Theorem 7.1. *The group*

$$\text{EL}_9(L_{\mathbb{F}_2}(1, 2))$$

is not sofic.

Proof. Let G, Γ, u, v, K , and J be as above. The groups G and Γ have property (T) by Theorem 4.2. Equations (37) and (38) give

$$G = \langle \Gamma, u, v \rangle, \quad u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma.$$

Proposition 6.2 gives

$$J \leq \Gamma, \quad [K, J] = 1, \quad K \cap J = 1.$$

Thus every hypothesis of Theorem 3.1 holds with $Q = \{u, v\}$ and $q_0 = u$. If G were sofic, that theorem would imply that J is LEF, contradicting Proposition 6.3. $\square$

Corollary 7.2. *The group $G = \text{EL}_9(R)$ is infinite, finitely generated, has property (T), and is nonsolic.*

Proof. Property (T) was established above, and a countable property-(T) group is finitely generated. Alternatively, finite generation follows directly because R is finitely generated as a ring and elementary commutator identities generate all ring coefficients from a finite ring generating set. The group is infinite because it contains $J \cong V$. Nonsolicity is Theorem 7.1. $\square$

8. A FINITELY PRESENTED NONSOFTIC GROUP

We finish with a general finite-table construction.

For a finite set $F \subseteq G$, a map

$$p : F \cup F^2 \longrightarrow \text{Sym}(Y)$$

is an (F, ε) -model if

$$(48) \quad d_H(p(g)p(h), p(gh)) < \varepsilon \quad (g, h \in F),$$

$$(49) \quad d_H(p(g), p(h)) > 1 - \varepsilon \quad (g, h \in F, g \neq h).$$

A countable group is sofic if and only if every finite F admits (F, ε) -models for every $\varepsilon > 0$.

Theorem 8.1 (Finite multiplication-table presentation). *Let G be a finitely generated nonsofic group. Then there is a finite symmetric set*

$$1 \in F = F^{-1} \subseteq G$$

containing a generating set of G such that the finitely presented group

$$(50) \quad H_F = \langle x_g \ (g \in F \cup F^2) \mid x_1 = 1, \quad x_g x_h = x_{gh} \ (g, h \in F) \rangle$$

is infinite and nonsofic.

Proof. Since G is nonsofic, there are a finite symmetric set F containing 1 and a finite generating set, and $\varepsilon_0 > 0$, for which no (F, ε_0) -model exists. The presentation (50) is finite. Evaluation of the labels defines a homomorphism

$$(51) \quad \pi : H_F \longrightarrow G, \quad \pi(x_g) = g.$$

Because F contains a generating set, π is surjective. Hence H_F is infinite.

Suppose that H_F were sofic. Choose a sufficiently accurate sofic approximation of a finite subset containing all x_g , $g \in F \cup F^2$, and all products and inverses needed below. Write $p(g)$ for the permutation assigned to x_g . Since

$$x_g x_h = x_{gh} \quad (g, h \in F)$$

holds exactly in H_F , approximate multiplicativity gives, after choosing the approximation accuracy small enough,

$$d_H(p(g)p(h), p(gh)) < \varepsilon_0 \quad (g, h \in F).$$

If $g \neq h$ in F , then $x_g \neq x_h$, since their images under π are distinct. Thus $x_g^{-1}x_h \neq 1$. Approximate multiplicativity and sofic faithfulness give

$$d_H(p(g), p(h)) = d_H(\text{id}, p(g)^{-1}p(h)) > 1 - \varepsilon_0$$

for a sufficiently accurate approximation. Hence p is a forbidden (F, ε_0) -model of G , a contradiction. $\square$

Proof of Theorem 1.1. Corollary 7.2 proves the first assertion. Apply Theorem 8.1 to that group to obtain H_F . $\square$

9. CONCLUDING REMARKS

Remark 9.1. The two compressors have different roles. Analytically, one compressor q_0 is enough to produce the matched component carrying the expanding K -action. Algebraically, the pair is useful because

$$vu^{-1} = z$$

is the block involution that generates all elementary matrices involving the ninth coordinate.

Remark 9.2. The value $1/2$ in the median argument is a normalization, not a corner trace. It records equality with the median size:

$$\frac{s}{s+m} = \frac{1}{2} \iff s = m.$$

The conservation identity is what converts the one-sided component comparison into two-sided size control.

Remark 9.3. The passage from G to H_F does not use closure of soficity under quotients. The surjection $H_F \twoheadrightarrow G$ is used only to prove infinitude and to distinguish the finitely many named generators. Nonsoficity is inherited from the explicit forbidden finite-table test.

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